

Degree Project in the Field of Technology Media Technology and the Main Field of Study Computer Science and Engineering

Second cycle, 30 credits

Beyond Accuracy: Evaluating Generative AI as Decision Support

A joint analysis of performance, confidence, and reliance in phishing detection

GRÖNSTEDT SÖDERBERG OLLE

Beyond Accuracy: Evaluating Generative AI as Decision Support

A joint analysis of performance, confidence, and reliance in phishing detection

GRÖNSTEDT SÖDERBERG OLLE

Master's Programme, Interactive Media Technology, 120 credits
Date: June 17, 2026

Supervisor: Haibo Li
Examiner: Anders Hedman
School of Electrical Engineering and Computer Science
Swedish title: Bortom träffsäkerhet: En utvärdering av generativ AI som beslutsstöd
Swedish subtitle: En samlad analys av prestation, beslutsäkerhet och tillit till AI vid identifiering av phishing

Beyond Accuracy: Evaluating Generative AI as Decision Support

A joint analysis of performance, confidence, and reliance in phishing detection

GRÖNSTEDT SÖDERBERG OLLE

Generative AI is increasingly used as decision support across multiple domains, yet its effects on performance, confidence, and reliance remain unclear, particularly when these dimensions are examined jointly. Prior research has often investigated such constructs separately and has primarily focused on traditional probabilistic AI systems rather than generative AI systems that provide recommendations and explanations in natural language. This study used a between-subjects experimental design to examine the effects of AI-supported decision-making in a phishing email detection task. Participants were assigned to either an AI-supported or control condition, and outcomes were analyzed. The results showed that AI support improved overall accuracy, primarily by reducing false negatives. No detectable difference in subjective confidence was observed between conditions. Participants also exhibited increased adherence to incorrect AI recommendations, revealing a pattern of partial-but-insufficient calibration to AI correctness. Overall, the findings suggest that AI decision support affects performance, confidence, and reliance differently, highlighting the importance of evaluating AI-supported decision-making across multiple dimensions rather than relying on a single outcome metric.

SAMMANFATTNING

Generativ AI används allt oftare som beslutsstöd inom flera olika områden, men dess påverkan på träffsäkerhet, beslutsäkerhet och benägenheten att följa AI-rekommendationer är fortfarande inte helt klarlagd. Tidigare forskning har ofta undersökt dessa dimensioner separat och främst fokuserat på traditionella sannolikhetsbaserade AI-system snarare än generativa AI-system som ger textbaserade rekommendationer. Studien använde en experimentell mellangruppsdesign för att undersöka hur AI-stöd påverkar beslutsfattande, i detta fall vid identifiering av phishingmejl. Deltagarna delades in i en AI-stödd grupp och en kontrollgrupp. Den AI-stödda gruppen fick stöd i form av korta AI-genererade klassificeringsförslag medan kontrollgruppen genomförde uppgiften utan AI-stöd. Studien undersökte specifikt påverkan på prestation, beslutsäkerhet och benägenheten att följa AI-rekommendationer. Resultaten visade att AI-stöd förbättrade den övergripande träffsäkerheten, främst genom att minska antalet falska negativa (missade phishingmejl). Däremot fann studien inga belägg för att AI-stödet påverkade deltagarnas subjektiva beslutsäkerhet. Deltagarna uppvisade samtidigt en ökad benägenhet att följa felaktiga AI-rekommendationer, vilket tyder på en delvis men otillräcklig förmåga att bedöma AI:ns korrekthet. Sammantaget tyder studiens resultat på att AI som beslutsstöd påverkar träffsäkerhet, beslutsäkerhet och benägenheten att följa AI-rekommendationer på olika sätt. Detta belyser vikten av att utvärdera AI-stött beslutsfattande utifrån flera olika dimensioner snarare än att förlita sig på ett enskilt mått.

Author's address: Grönstedt Söderberg Olle, ollegs@kth.se, KTH Royal Institute of Technology, School of Electrical Engineering and Computer Science, Stockholm, Sweden, SE 100 44.

Document date: June 17, 2026

Swedish title: Bortom träffsäkerhet: En utvärdering av generativ AI som beslutsstöd

Swedish subtitle: En samlad analys av prestation, beslutsäkerhet och tillit till AI vid identifiering av phishing

© 2026 Copyright held by the author.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; • **Computing methodologies** → *Artificial intelligence*; • **Security and privacy** → Phishing.

Keywords: generative AI, decision support, phishing detection, reliance, confidence, human-AI decision-making

Nyckelord: generativ AI, beslutsstöd, nätfiske, beslutsäkerhet, AI-stöd, människa-AI-beslutsfattande

1 INTRODUCTION

1.1 Generative AI as Decision Support

Generative AI systems such as ChatGPT are increasingly used as decision-support tools across professional and organizational contexts [6, 9, 11, 16, 17]. In the present study, the term generative AI refers specifically to large language model (LLM)-based conversational systems that generate text-based recommendations and explanations. Research on such systems is particularly common in medicine, education, and military contexts, but also covers broader decision-making situations such as general trivia. Prior research suggests that AI support can improve task performance in certain domains [2, 14]. At the same time, studies indicate a risk for automation-bias, and that AI support may influence users' confidence and reliance behavior [4, 7].

As these systems become embedded in everyday work practices, and function as collaborators rather than mere tools, understanding these dynamics becomes increasingly critical, as decisions made with AI support not only appear more often, but may carry legal, operational, financial, or security-related consequences.

Research on AI-assisted decision-making has examined a variety of performance and behavioral metrics. For this study, three constructs are particularly relevant: performance, confidence, and reliance. Examining these constructs in isolation risks generating misleading insights. For instance, a study measuring only accuracy might suggest that AI is clearly beneficial, but reliance analysis could reveal that users are following incorrect AI recommendations more often which complicates that conclusion. Conversely, a study focused on reliance might suggest overreliance is a problem, but accuracy analysis could reveal that the net performance gain still outweighs the cost. Analyzing the three constructs jointly makes such trade-offs visible, which is the approach taken in the present study.

1.2 Research Questions and Hypotheses

MRQ: How does generative AI-based decision support influence accuracy, confidence, and reliance in a risk-sensitive classification task?

RQ1: How does AI-support influence decision accuracy and error types compared to unaided human decision-making?

H1: AI-supported participants will demonstrate significantly higher decision accuracy compared to participants without AI support.

RQ2: How does AI-support influence subjective confidence in decision-making?

H2: Participants receiving AI-support will report significantly higher decision confidence than participants in the control condition.

RQ3: To what extent do participants rely on AI recommendations, and is such reliance calibrated to the AI's actual correctness?

H3: Participants will follow AI recommendations significantly more often when the AI is correct than when it is incorrect.

2 LITERATURE REVIEW METHOD

The literature review was conducted to identify prior research on AI-assisted decision-making, with particular focus on three constructs central to the present study: objective task performance, decision confidence, and reliance behavior. The process was informed by recommendations for conducting literature studies in information systems and computing research outlined by Oates [12]. Relevant studies were identified primarily through searches in Web of Science and the digital library services provided by KTH and Uppsala University.

Search queries included combinations of keywords: AI decision support, automation bias experiment, trust in automation and decision-making, decision support experiment, and ChatGPT phishing detection experiment. Additional literature was identified through backward citation tracking of frequently cited papers in the field.

Priority was given to peer-reviewed journal articles and conference proceedings within human–computer interaction, decision science, and AI research communities. Given the rapidly evolving nature of generative AI, relevant recent preprints were also considered where appropriate. While the primary focus of this study is phishing detection, literature from other high-stakes decision-making domains (e.g., medical diagnosis, military decision support) was also considered, as these involve comparable decision structures with significant consequences for incorrect judgments.

3 THEORETICAL BACKGROUND

3.1 AI-Assisted Decision-Making

AI-assisted decision-making has emerged as a growing area of research as AI systems are increasingly integrated into professional workflows. Existing work covers medicine [8, 15, 17], where AI is used to support diagnostic reasoning and treatment recommendations; education [4, 8], where AI can be used to provide feedback or tutor learners; and various IT related domains [5, 13], such as cybersecurity and software development tasks. A recurring concern in this literature is whether AI support improves objective decision performance or instead introduces risks related to miscalibrated confidence and overreliance on AI recommendations. Three constructs are especially prominent in this literature: objective task performance, subjective decision confidence, and behavioral reliance on AI recommendations. These have been identified as central dimensions in empirical research on human–AI decision-making [8] and together serve as the analytical lens for the present study. The following sections review existing findings related to each of these constructs.

3.2 Performance

Recent work has begun to examine how AI assistance affects objective task performance [10, 14]. Although findings vary across contexts and study designs, a substantial portion of empirical research suggests that integrating AI into decision-making processes can increase the likelihood of improved objective task performance, such as higher accuracy or more satisfactory outcomes [3, 4, 8, 13]. At the same time, research has also indicated that performance gains are context-dependent; for example, user expertise and task complexity may moderate the extent to which AI support leads to improved outcomes [8, 13, 17]. AI support may also affect error types differently, reducing some while increasing others [13]. In practice, the importance of these error types depends on the consequences associated with them. For example, in security-sensitive tasks such as phishing detection, a false positive (incorrectly classifying a legitimate email as phishing) is generally preferable to a false negative (incorrectly classifying

a phishing email as legitimate), as missed phishing emails can lead to security breaches, whereas false positives typically result in minor inconvenience. Distinguishing between these errors is therefore important, since different error types can have very different practical consequences depending on the context. Yet even when error types are considered, accuracy alone does not provide a complete picture of how AI affects decision-making.

3.3 Confidence

Decision confidence is also commonly researched in relation to human-AI decision-making and refers to the degree to which a person believes their decision is correct. Given the conversational nature of generative AI systems, which communicate recommendations through natural language rather than explicit probability estimates, the support they provide often shapes users' confidence assessments, making confidence a relevant metric for investigation [16].

Based on the reviewed literature, AI support has been shown to increase users' subjective confidence, sometimes beyond what is warranted by objective performance [10, 13, 16]. In addition to overall confidence levels, confidence calibration reflects the alignment between confidence and actual correctness and has been linked to decision quality, as it can reveal whether decision-makers are overconfident in incorrect judgments or insufficiently confident in correct ones [10].

However, decision confidence appears complex, as it has been shown to vary depending on domain expertise [4, 10], AI literacy [8] and task complexity [11]. Importantly, confidence captures only the subjective dimension of the decision and does not reveal whether individuals arrived at their choice independently or by following the AI recommendation. This behavioral dimension is captured by reliance.

3.4 Reliance

Lastly, Reliance behavior refers to the extent to which individuals adopt or override AI recommendations when making a final decision. For this construct, existing papers often discuss overreliance (following incorrect AI advice) and underreliance (rejecting correct AI advice) [4, 8, 10, 11]. Many of which report a tendency toward overreliance, where users follow incorrect AI recommendations, although the concept is operationalized differently across studies [4, 8, 10]. While over- and underreliance are not inherently good or bad, they provide a useful basis for evaluating how individuals respond to AI recommendations. Evaluating reliance relative to AI correctness helps explain whether users follow correct advice and override incorrect advice, and thus whether their reliance behavior is calibrated to the AI's performance. Such calibration may be partial; prior work suggests that while individuals adjust their behavior in response to AI correctness, this adjustment is often insufficient to fully align reliance with the AI's actual performance. A related problem is how overreliance is measured. Studies typically count how often participants follow incorrect AI advice, but without comparing to participants doing the same task without AI, it is unclear whether those errors were caused by the AI or simply by the difficulty of the task [1]. Prior work also suggests that reliance behavior is affected by the length of the AI response. [16] found that participants were more likely to follow AI recommendations when the AI provided longer and more detailed explanations, referring to this phenomenon as a "length bias" [16].

3.5 Research Gap and the Present Study

Generative AI. A substantial amount of research on AI-supported decision-making focuses on classical or probabilistic machine learning systems that communicate recommendations through explicit

numerical confidence estimates [4, 8, 10, 18]. Generative AI tools differ in how they communicate recommendations, presenting them through natural language and reasoning rather than calibrated probability scores. Because users can in principle assess the reasoning behind a recommendation rather than relying solely on a numeric score, the mechanisms shaping confidence and reliance may operate differently. As a result, it remains unclear whether findings from classical AI systems generalize to this context, highlighting a gap that warrants empirical investigation using *generative* AI as decision support.

Isolated Metrics. While prior research has examined performance, confidence, and reliance, these metrics are frequently investigated in isolation [8, 10]. Individually, each metric offers a limited view: high accuracy may appear reassuring, but if it is driven by uncritical adherence to AI recommendations, the same behavior that improves performance on correct AI outputs could also increase errors when the AI is wrong. Similarly, unchanged confidence can be misleading if considered on its own, as it does not reveal whether decisions are made independently or influenced by AI recommendations [10]. Importantly, these constructs are interrelated and capture complementary aspects of AI-assisted decision-making [10]. Yet they are rarely examined jointly within the same decision task, leaving the trade-offs between performance, confidence, and reliance underexplored.

To address these gaps, the present study examines all three constructs jointly in a single generative-AI-supported classification task in a security-sensitive setting. By doing so, it provides empirical evidence on how natural-language AI recommendations shape the interplay between objective performance, subjective confidence, and behavioral reliance.

4 METHODOLOGY

4.1 Research Design

The present study adopted a quantitative research design using a controlled between-subjects experiment with two conditions: an AI-supported group and a control group receiving no additional support. A between-subjects design was selected to avoid the carry-over and task-learning effects that could arise in a within-subjects design, where exposure to the same stimuli under both conditions could allow participants to learn recurring phishing cues and distort measures of accuracy, confidence, and reliance.

The experiment was implemented through a survey-based interface. Participants were presented with identical email stimuli and asked to classify each as phishing or legitimate, while also reporting their confidence in each decision. This design enabled systematic manipulation of the independent variable, namely the presence or absence of AI support, while holding all other factors constant. Decision accuracy, confidence, and reliance behavior were measured as dependent variables.

Experiments are well suited to investigating relationships between independent and dependent variables, particularly when causal inference is of interest [12], which made the design appropriate for isolating the effect of AI support.

4.2 Participants

Participants were mainly recruited from four different companies in Sweden that agreed to participate in and distribute the study internally. Recruitment was conducted via designated contact persons at each organization, who shared the survey link with employees.

To supplement this, additional participants were recruited through extended professional and personal networks, using purposive, snowball, and convenience sampling [12].

Participation was open to individuals aged 18 and above who regularly use email in a work-related context. While no formal verification of age was conducted, participants were primarily recruited through professional environments, where all employees are expected to meet this criterion.

This broad inclusion was intentional, as the study aims to reflect real-world decision-making contexts, where individuals with diverse backgrounds and varying levels of experience interact with email and AI-supported tools.

4.2.1 Sample Size. An a priori power analysis was conducted based on a medium expected effect size (d = 0.50). This assumption was mainly informed by [11], who applied a similar effect size in an experimental comparison of AI-supported and non-AI-supported task performance.

Using an independent samples t-test with $\alpha = .05$ and desired statistical power of .80, the analysis indicated a target sample size of approximately 128 participants. The final sample, however, consisted of 81 participants in total, distributed across the two conditions; implications of this discrepancy for statistical power are discussed in Section 8.

4.2.2 Ethical considerations. The study involves a phishing detection task, but all stimuli consist of simulated email examples that do not contain active links or any elements that could cause harm.

Participation was voluntary, and participants could withdraw at any time without consequence. Prior to participation, individuals received information about the study and provided informed consent.

No personally identifiable information was collected. All responses were stored securely in a restricted-access cloud storage environment and were only accessible to the author of the study and, if necessary, the academic supervisor or examiner. Furthermore, since the study builds on stimuli generated by AI, it could be argued to have a negative environmental impact, which is worth acknowledging.

4.3 Pilot Study

A pilot study was conducted with five participants to evaluate the clarity, realism, and difficulty of the stimuli, as well as the overall feasibility of the experiment itself.

The pilot results indicated that the estimated completion time of 10–15 minutes was appropriate. Participant responses showed a varied distribution across stimuli, suggesting that the task was neither trivial nor overly difficult. The predefined difficulty gradient was also reflected in the results, where easy stimuli were more frequently classified correctly than medium and hard stimuli. In addition, responses to ambiguous legitimate emails were more evenly distributed, which was expected given their intended design.

However, two medium-difficulty phishing emails (#4 and #23) were found to be more difficult than intended. These stimuli were subsequently revised to better align with the intended difficulty level. Given the small pilot sample, these patterns were used for light calibration rather than major redesign.

In addition to participant-based pilot testing, domain-expert feedback was collected from two professionals working in IT security. They were asked to evaluate the realism of the emails, identify any inconsistencies, and comment on whether the scenarios reflected plausible workplace situations. This feedback indicated that the emails were generally perceived as realistic, and led to minor adjustments to improve clarity and consistency, while preserving intentionally designed ambiguities.

Overall, the pilot study supported the validity of the stimuli and confirmed that the experimental design functioned as intended.

4.4 Experimental Task

Participants were allocated to either the AI-supported or control condition using a predefined assignment rule based on birth month (odd-numbered months vs. even-numbered months). Due to limitations of the selected survey tool, true randomization could not be implemented directly within the survey flow. Although this approach does not constitute true randomization, it was considered unlikely to introduce systematic bias as there is no clear theoretical basis for birth month being associated with phishing detection ability.

All participants completed a total of 24 trials through a digital survey interface. In each trial, the participants were presented with a single email stimulus. The participants were then asked to read the email and classify it as either phishing or legitimate and to indicate their confidence in the decision. In the AI-supported condition, each email was accompanied by an AI-generated recommendation and a short standardized explanation.

The task was self-paced, and participants were informed that it would take approximately 10-15 minutes to complete. The order of the email stimuli was fixed and identical across all participants.

4.4.1 Stimuli. In this study, phishing emails are defined as deceptive messages designed to impersonate a trusted entity or context in order to induce the recipient to perform actions such as revealing credentials, clicking a link, downloading an attachment, providing sensitive information, or similar behaviors.

The stimuli were evenly divided between phishing and legitimate emails. The difficulty of the stimuli varied in order to reduce the risk of ceiling effects (i.e., participants achieving near-perfect scores) and to capture a range of decision-making scenarios. Legitimate and phishing emails were categorized separately: legitimate emails were classified as either clear or ambiguous (based on how easily their legitimacy could be determined), while phishing emails were classified as easy, medium, or hard (based on how noticeable the phishing cues were). Difficulty assignments were based on the salience and number of phishing cues. Easy emails contained conspicuous red flags visible at first glance, medium emails had subtler cues that required closer reading, and hard emails required contextual inference. Existing phishing examples were identified through online searches, publicly available examples, and common impersonation patterns described in cybersecurity literature and observed in real-world contexts. Depending on their suitability for the experimental setting, examples were either used directly or modified to better align with the intended format, realism, and difficulty level of the study. Similarly, legitimate emails were modelled on common workplace communication patterns, such as routine notifications, internal updates and requests, and standard external correspondence.

To further ensure realism, the design of the email stimuli was also informed by practitioner input collected from employees across all four participating organizations. This input was gathered through short written responses and follow-up discussions, including email correspondence and video calls. Employees were asked to describe common indicators of phishing emails, situations in which phishing is difficult to detect, and factors that signal legitimacy. The feedback highlighted recurring patterns such as suspicious sender addresses, language inconsistencies, urgent requests, and unexpected attachments or links. These insights were then used to shape the stimuli and ensure that the emails reflected patterns commonly encountered in real-world phishing scenarios.

The stimuli were designed in English to preserve realism and consistency with common real-world examples, many of which originate from English-language contexts (e.g., system notifications and phishing

templates). Translating such messages into Swedish could introduce unintended phrasing or cues that influence participants' judgments. Additionally, the design focused primarily on email content rather than visual detail. While basic formatting was included to support realism, elements such as logos, exact fonts and branding were not emphasized, as excessive visual detail could introduce unintentional factors unrelated to the classification task.

The number of stimuli (24) was chosen to provide sufficient variation across trials while keeping the task within the 10–15 minute duration established in the pilot study. This number also allowed for an even split between phishing and legitimate emails, with multiple difficulty levels represented within each category. To enable analysis of user reliance, six of the 24 AI recommendations (25%) were intentionally assigned incorrect classifications ("seeded errors"). These errors included both false positives and false negatives and were distributed across difficulty levels while maintaining a plausible overall AI accuracy level (75%). Examples of stimuli across each difficulty category, including the AI recommendations shown to participants, are provided in Appendix B.

4.4.2 AI Support. In the AI-supported condition, each email was accompanied by an AI-generated classification (phishing or legitimate) along with a brief natural-language explanation. To ensure that differences between conditions could be attributed to the presence or absence of AI support rather than variability in AI responses, all AI outputs were standardized and pre-generated for each stimulus, ensuring that all participants received identical recommendations.

All outputs were generated using ChatGPT (GPT-5.3, OpenAI) in April 2026. A consistent prompt structure was applied across all stimuli, instructing the model to provide a categorical classification and a concise explanation, without including numeric confidence scores (see Appendix A for full prompt). This design choice was made to avoid influencing participants' judgments through explicit probability cues and to better reflect how generative AI systems typically communicate recommendations through natural language. Furthermore, explanations were deliberately kept short (1–2 sentences) and similar in length across stimuli. As previously mentioned, prior work has identified explanation length as a factor that can influence reliance and confidence [16]. Holding explanation length relatively constant across trials therefore helped ensure that observed differences were more directly related to the presence of AI support.

To limit contextual carryover effects from prior interactions or session-specific settings, each stimulus was processed in a separate incognito browser session without an active login. Additionally, all prompts were generated in a single sitting to minimize potential time-related variability in model responses, as suggested in prior work [5].

4.5 Procedure

The experiment was conducted as an online survey. Participants received a recruitment message containing two survey links, one for each condition, and selected the appropriate link based on their birth month. Upon entering the survey, participants provided informed consent and received task instructions before proceeding to the classification trials themselves.

4.6 Measures

To capture different aspects of AI-supported decision-making, measures were defined across the three constructs identified in Section 3 as central to human-AI decision-making: performance, confidence,

and behavioral reliance. These constructs correspond to the study's research questions, capturing objective performance (RQ1), subjective confidence (RQ2), and behavioral reliance on AI recommendations (RQ3).

4.6.1 Performance Measures. Decision performance was primarily assessed using classification *accuracy*, calculated per participant as the proportion of correctly classified emails relative to the total number of trials and compared between conditions:

$$\textit{Accuracy} = \frac{\text{Number of correct classifications}}{\text{Total number of trials}}$$

In addition to overall accuracy, *error types* were analyzed to provide further insight into participants' decision behavior. Incorrect classifications were categorized as false positives (legitimate emails incorrectly classified as phishing) and false negatives (phishing emails incorrectly classified as legitimate). These were calculated as proportions relative to the total number of relevant trials:

$$\textit{False Positive Rate} = \frac{\text{Legitimate emails classified as phishing}}{\text{Total legitimate emails}}$$

$$\textit{False Negative Rate} = \frac{\text{Phishing emails classified as legitimate}}{\text{Total phishing emails}}$$

4.6.2 Confidence Measures. *Subjective confidence* was measured using self-reported ratings on a 7-point Likert scale for each trial. Mean confidence was calculated per participant and compared between conditions.

To assess the relationship between confidence and correctness, the confidence-accuracy gap was examined. This was operationalized as the difference in mean confidence between correct and incorrect trials. This measure indicates whether higher confidence is associated with correct decisions and whether AI support affects this relationship.

4.6.3 Behavioral Measures. In the AI-supported condition, behavioral reliance on AI recommendations was assessed through *adherence rates*. Overall adherence was defined as the proportion of trials in which the participant's classification matched the AI recommendation:

$$\textit{Adherence} = \frac{\text{Trials where participant agrees with AI}}{\text{Total trials}}$$

To assess whether reliance was appropriately calibrated, adherence was further analyzed separately for trials where the AI was correct and where it was incorrect:

$$\textit{Adherence}_{\text{correct}} = \frac{\text{Agreement on trials where AI is correct}}{\text{Total trials where AI is correct}}$$

$$\textit{Adherence}_{\text{incorrect}} = \frac{\text{Agreement on trials where AI is incorrect}}{\text{Total trials where AI is incorrect}}$$

Comparing these measures provides insight into whether participants selectively rely on AI when it is accurate, or whether they exhibit overreliance or underreliance.

5 DATA ANALYSIS PLAN

5.1 Pre-processing

As all survey questions were mandatory, no incomplete trial-level responses occurred, and all data were included as recorded. Trial-level responses were then compiled into participant-level metrics, including overall accuracy, mean confidence, and adherence rates, which were then used for subsequent group-level comparisons.

5.2 Descriptive and inferential analysis

The present study used descriptive statistics to summarize the central tendencies and variability of the collected data across experimental conditions. Inferential statistical analyses were also conducted to evaluate differences between conditions and to test the hypotheses associated with each research question. All statistical analyses were conducted with Python using the SciPy and statsmodels libraries, with pandas and NumPy used for data processing.

Normality was assessed using the Shapiro-Wilk test. If Shapiro-Wilk indicated approximate normality in both groups ($p > .05$), Welch's independent-samples t-tests were used, otherwise the Mann–Whitney U test was applied. For paired comparisons, the same logic was applied based on the distribution of difference scores, using either paired-samples t-tests or Wilcoxon signed-rank tests. All tests were two-tailed with $\alpha = .05$, and effect sizes (Cohen's d for t-tests, Cohen's dz for paired t-tests, or rank-biserial r for non-parametric tests) are reported alongside p-values.

5.2.1 Analysis of RQ1. Decision accuracy and error rates were calculated for each participant. Differences between the AI-supported and control groups were examined using the between-group statistical comparisons described in Section 5.2.

5.2.2 Analysis of RQ2. Mean confidence ratings were calculated per participant and compared between the AI-supported and control conditions using between-group statistical comparisons. In the case of a non-significant primary comparison, a Two One-Sided Tests (TOST) equivalence test with bounds of $\pm d = 0.30$ was conducted to assess whether the conditions could be considered practically equivalent in terms of mean confidence. As a secondary analysis, confidence was also examined separately for correct and incorrect trials to assess whether AI support affects the calibration of confidence to actual performance.

5.2.3 Analysis of RQ3. Adherence to AI recommendations was calculated for participants in the AI-supported condition. Adherence rates for trials where the AI was correct and incorrect were compared using within-participant statistical comparisons (a paired samples t-test) to assess whether reliance was calibrated to AI performance. In addition, the rate at which AI-supported participants followed incorrect AI recommendations was compared with the rate at which participants in the control condition made the same classifications independently. This comparison was used to distinguish AI-induced overreliance from errors simply caused by the inherent difficulty of the task.

5.2.4 Secondary analysis. As the stimuli were designed along a predefined difficulty gradient (Section 4.4.1), a secondary analysis examined whether AI support had different effects across difficulty

levels. This analysis was planned as part of the design but was not associated with a directional hypothesis.

5.3 Validity Considerations

By presenting participants with identical stimuli and ensuring that AI support constituted the primary difference between conditions, observed differences can more confidently be attributed to the presence of AI support. While this supports internal validity, condition assignment was based on birth month rather than true randomization, which introduces a potential risk of uneven group composition. Although no clear mechanism suggests that birth month is associated with the measured constructs, this approach may still introduce uncontrolled variation across conditions. Additionally, as the experiment was conducted online, participants completed the task in uncontrolled environments, introducing potential variation in external conditions. Lastly, one should acknowledge that learning or fatigue across the email classifications is an inherent risk of the study design.

The use of phishing email classification provides a controlled and realistic task context. However, it represents a specific type of decision-making scenario and may not fully capture the complexity of broader real-world decision processes. A key limitation is that adherence is used as a proxy for reliance. It does not distinguish between participants who independently reached the same conclusion as the AI and those who simply followed its recommendation. To address this, the study compared how often AI-supported participants agreed with incorrect AI recommendations to how often the control group produced the same incorrect classification without AI. This shows how much of the agreement was caused by the AI, rather than participants simply making the same mistake regardless.

External validity is influenced by the experimental setup. While the use of phishing email classification provides a realistic task, the study was conducted in a controlled survey environment, which differs from real-world email contexts where decisions are made under varying conditions and constraints.

6 RESULTS

6.1 Descriptive results

Table 1 presents the descriptive statistics for all metrics examined in the present study. Overall, accuracy was higher in the AI-supported condition than in the control condition. This difference was primarily driven by a lower false negative rate (fewer missed phishing emails). Confidence levels were nearly identical across conditions. Within the AI-supported condition, adherence was substantially higher on trials where the AI was correct than on trials where it was incorrect. The following sections report inferential analyses for each research question.

6.2 RQ1: Accuracy and Error Rates

Accuracy was higher in the AI-supported group (M = 0.75, SD = 0.10) compared to the control group (M = 0.69, SD = 0.11). As normality assumptions were violated, a Mann–Whitney U test was conducted, indicating a statistically significant difference between groups (U = 1096.0, p = .009, r = −.34, 95% CI [−0.52, −0.13]).

The *False Positive Rate* was slightly higher in the AI-supported group (M = 0.26, SD = 0.24) than in the control group (M = 0.22, SD = 0.22). As normality assumptions were violated, a Mann–Whitney U test was conducted and indicated that this difference was not statistically significant (U = 984.0, p = .114, r = −.20, 95% CI [−0.40, 0.02]).

Table 1: Descriptive Statistics for Accuracy, Confidence, and Adherence Across Conditions

Measure	AI Condition (n = 40)	Control (n = 41)
Accuracy		
Overall	0.75 (0.10)	0.69 (0.11)
False positive rate	0.26 (0.24)	0.22 (0.22)
False negative rate	0.24 (0.13)	0.41 (0.21)
Confidence		
Mean confidence	5.33 (0.77)	5.42 (0.72)
Adherence (AI condition only)		
AI correct trials	0.85 (0.13)	—
AI incorrect trials	0.58 (0.23)	—

Note. Values are presented as M (SD).

The *False Negative Rate* was lower in the AI-supported group (M = 0.24, SD = 0.13) compared to the control group (M = 0.41, SD = 0.21). As normality assumptions were met, a Welch's t-test was conducted, revealing a statistically significant difference (t = −4.28, p = .0001, d = −0.95, 95% CI [−1.41, −0.49]).

These results partially support H1: AI-supported participants demonstrated significantly higher overall accuracy, primarily driven by a reduction in false negatives. The false positive rate did not differ significantly between conditions.

6.3 RQ2: Confidence

Mean confidence did not differ significantly between the AI-supported group (M = 5.33, SD = 0.77) and the control group (M = 5.42, SD = 0.72). A Mann–Whitney U test indicated no significant difference (U = 755.5, p = .545, r = .08, 95% CI [−0.14, 0.29]). A TOST equivalence test with bounds of ±d = 0.30 was also non-significant (p = .200), meaning equivalence could not be formally established.

As a secondary analysis, confidence was examined separately for correct and incorrect trials. No significant differences between conditions were found for confidence on correct trials (d = −0.06, p = .69), confidence on incorrect trials (d = −0.36, p = .109), or the confidence-accuracy gap (d = 0.39, p = .085).

H2 was not supported: AI support did not significantly increase subjective confidence.

6.4 RQ3: Reliance behavior

Within the AI-supported condition, participants adhered to AI recommendations significantly more often when the AI was correct (M = 0.85, SD = 0.13) than when it was incorrect (M = 0.58, SD = 0.23). As the difference scores were approximately normally distributed (Shapiro-Wilk, W = 0.97, p = .39), a paired-samples t-test was conducted, confirming this difference (t = 7.46, p < .0001, dz = 1.18, 95% CI [0.78, 1.58]).

The between-group comparison on seeded-error trials showed that AI-supported participants matched the incorrect AI recommendation at a significantly higher rate (M = 0.58, SD = 0.23) than control participants independently gave the same incorrect classification (M = 0.39, SD = 0.18). A Mann–Whitney U test indicated a significant difference (U = 1203.0, p = .0002, r = −.47, 95% CI [−0.62, −0.28]).

H3 was supported: participants followed AI recommendations significantly more often when the AI was correct. However, the cross-condition comparison indicates that AI support also increased the rate of incorrect classifications on seeded-error trials compared to unaided decision-making.

6.5 Secondary Analysis Results: Difficulty

As a secondary analysis, accuracy was compared between conditions across difficulty levels. AI support was associated with significantly higher accuracy on Medium-difficulty phishing emails (AI: M = 0.70, Control: M = 0.39; U = 1270.5, p < .001, r = –.55) and Hard phishing emails (AI: M = 0.61, Control: M = 0.47; U = 1056.5, p = .021, r = –.29). No significant differences were observed for Clear, Easy, or Ambiguous stimuli, although the Ambiguous category showed a non-significant trend (p = .056) in which AI-supported participants performed slightly worse than control participants. Given the exploratory nature of this analysis and the number of comparisons conducted, results should be interpreted with caution due to the increased risk of Type I error.

Table 2: Accuracy on Phishing Emails by Difficulty Level Across Conditions

Difficulty	AI (n = 40)	Control (n = 41)	p
Easy	0.96 (0.11)	0.91 (0.20)	.364
Medium	0.70 (0.24)	0.39 (0.31)	$<$.001
Hard	0.61 (0.21)	0.47 (0.31)	.021

Note. Values are mean accuracy (SD). Rank-biserial effect sizes: Medium $r = -.55$, Hard $r = -.29$; Easy was non-significant.

7 DISCUSSION

The study examined how generative AI-based decision support influences performance, confidence, and reliance in a phishing classification task. The results show that AI support significantly improved overall accuracy, primarily by reducing false negatives, while confidence remained largely unchanged across conditions. At the same time, reliance analysis indicated that participants frequently followed AI recommendations, including when those recommendations were incorrect, suggesting that the observed accuracy gains are partly dependent on the AI's own correctness. Together, these findings point to a shift in decision behavior that is not reflected in users' reported confidence.

RQ1. H1 was partially supported. The improvement in *accuracy* was driven almost entirely by a reduction in false negatives (fewer missed phishing emails). The mean false negative rate dropped from 41% in the control group to 24% with AI support, while false positives remained largely unchanged. This indicates that AI support made participants better at catching threats, but did not substantially improve their ability to correctly classify legitimate emails. One possible explanation is that AI support may have directed participants' attention toward potentially suspicious elements within the email, making threat-related cues more salient and thereby reducing the likelihood of overlooking phishing attempts.

The unchanged false positive rate may have a different explanation. Phishing emails tend to share identifiable patterns such as urgency, impersonation, or requests for sensitive information. Although the stimuli in this study were designed to vary in cues, structure, and difficulty, the AI was able to identify these recurring patterns with some consistency. Legitimate emails, by nature, vary widely in both content and structure, making it more difficult to provide conclusive evidence of legitimacy. Detecting the presence of threat indicators may therefore be more straightforward than confirming their absence, which could explain why AI support improved phishing detection without a corresponding improvement in correctly classifying legitimate emails.

Consistent with prior research, AI support improved overall decision accuracy [4, 13]. However, [13] found that AI support in a phishing detection task primarily reduced false positives. In contrast, the improvement in the present study was driven by a reduction in false negatives, with no significant change in false positives, indicating a different error pattern. One possible explanation lies in differences in unaided baseline performance. [13] used easy phishing trials that participants rarely missed, leaving little room for AI support to reduce false negatives further. The present study, by contrast, included a deliberate difficulty gradient with easy, medium, and hard phishing emails, where unaided false-negative rates were substantially higher, leaving greater room for AI support to improve performance. Although [13] exploratory focus-group design differs methodologically from the present experiment, their study remains a relevant point of comparison, as one of the few examining generative AI support in phishing detection. Interestingly, this room-to-improve pattern is reinforced by the secondary difficulty analysis, where AI support produced significant accuracy gains specifically on medium- and hard-difficulty phishing emails, the categories where unaided participants struggled most.

However, accuracy alone does not reveal how this improvement was achieved. The observed gains may partly result from participants simply following AI recommendations rather than independently improving their phishing detection performance, an interpretation explored further in the RQ3 discussion.

RQ2. H2 was not supported, as mean confidence ratings were essentially identical across the conditions (AI: M = 5.33, SD = 0.77; Control: M = 5.42, SD = 0.72). A TOST equivalence test was also non-significant, meaning that practical equivalence between the groups could not be formally established. Taken together, these results suggest that the presence of AI support did not produce a detectable change in participants' confidence. However, the inconclusive equivalence test means that a small effect cannot be entirely ruled out either.

The secondary analysis comparing *confidence* on correct and incorrect trials, as well as the confidence–accuracy gap, also showed no significant differences between the conditions. However, descriptively, the AI group appeared somewhat less confident on trials they classified incorrectly, resulting in a larger confidence–accuracy gap compared to the control group. These differences were not statistically significant and should therefore be interpreted with caution; that said, the pattern points toward better alignment between confidence and accuracy, rather than higher confidence overall.

Prior work on AI-supported decision-making suggests that AI assistance can inflate users' confidence beyond what is warranted by objective performance [16] (see also [11]). However, the present study did not replicate this pattern. This may be explained by the difference in the AI recommendations themselves. [16] identify explanation length as a key driver of confidence inflation, where longer explanations lead to higher user confidence even when they convey no additional information. In contrast, the AI recommendations in the present study were kept short and standardized (1–2 sentences) for experimental control. Hence, by keeping explanations brief and uniform, the present study plausibly removed the cue that drives confidence inflation, which could explain why subjective confidence remained unchanged despite the observed gain in accuracy.

RQ3. H3 was supported. Participants adhered significantly more to correct AI recommendations (85%) than incorrect AI recommendations (58%), showing a large effect ($d_z = 1.18$, $p < .0001$). In practical terms, when the AI was right participants followed it 85% of the time, but when it was wrong they followed it only 58% of the time which is a substantial drop. This difference in adherence might suggest appropriate calibration to AI correctness. However, the seeded-error comparison (the six emails with

intentionally incorrect recommendations) reveals the limits of this difference. On the same trials, AI participants matched the wrong AI answer 58% of the time, compared to 39% in the control condition ($p = .0002$, $r = -.47$). That is approximately an 18-percentage-point increase in error rate that can be attributed to the presence of AI support. Together, these findings show that participants followed correct recommendations more often than incorrect ones, but still followed incorrect recommendations often enough to produce AI-induced overreliance.

Considered separately, the two *reliance* analyses tell different stories. The first examined whether participants followed AI recommendations differently depending on whether the AI was correct. Adherence was substantially higher on correct AI trials, which on its own could suggest appropriate calibration to AI correctness. The second analysis focused on the seeded-error trials, examining whether AI-supported participants were more likely to give incorrect answers than control participants who received no AI support. This was indeed the case. Taken on its own, this pattern could suggest overreliance on AI recommendations. As outlined in Section 3.5, the three-construct design of the present study was intended to enable exactly this kind of joint analysis, and the combined findings fit neither reading alone. Instead, they point to partial calibration with insufficient correction: participants adjusted their behavior depending on AI correctness to some extent, but not enough to cancel out the influence of the incorrect recommendation.

Previous work on AI and overreliance falls along a spectrum. For instance, [7] found that the simple awareness that advice was AI-generated was enough to cause overreliance, even when the advice contradicted participants' own assessments and the available context. Others emphasize a partial calibration, where people do adjust to AI correctness, but they do it imperfectly. For example, [4] found that textual AI explanations made participants more likely to follow the AI when it was correct, but also more likely to follow it when it was wrong. [10] reported a similar asymmetry, where participants became more likely to follow correct AI recommendations while still remaining susceptible to incorrect ones. As shown above, the two reliance analyses point toward different interpretations when considered in isolation. The adherence analysis aligns with a partial-calibration perspective, as participants followed the AI substantially more when it was correct. In contrast, the seeded-error comparison aligns more closely with an overreliance perspective, as AI-supported participants still adopted the incorrect recommendations significantly more often than control participants gave the same answer on their own. Considered together, the findings suggest that participants can exhibit both partial calibration and overreliance simultaneously.

Existing overreliance studies have a measurement problem. These studies commonly measure how often participants follow incorrect AI advice. However, that number alone does not indicate whether the participants would have made the same errors without the AI due to the difficulty of the task. To determine that, some form of comparable baseline is required [1]. The present study provides such a baseline by comparing AI-supported and control participants on the same trials, meaning that task difficulty was held constant across conditions. The resulting 18-percentage-point gap in the rate of giving the wrong answer on these trials can therefore be more directly attributed to the presence of AI support.

Considered alongside the RQ1 finding that AI support produced a net accuracy increase, it is clear that this gain came with a cost. The same support that improved overall accuracy also introduced errors on trials where the AI was wrong, errors that would not have occurred without it. In the context of the given task (classifying phishing emails), a net accuracy gain is the desirable outcome. However, the trade-off still matters because the consequences of AI-induced errors are not symmetric. A false negative caused

by following incorrect AI advice could in practice mean a successful breach. In short, the net gain is good, but the induced errors are not free. Depending on the task and potential consequences, they may warrant additional safeguards in similarly high-stakes domains.

MRQ. Returning to the main research question - how generative AI-based decision support influences accuracy, confidence, and reliance jointly - it is evident that each construct, examined alone, supports a different conclusion. RQ1 alone would suggest that AI is a clear performance win. RQ2 alone would suggest no detectable change in confidence. RQ3 alone would suggest that AI causes overreliance. The joint picture shows that while all three are partially true, the trade-offs between them are what determine whether AI support is beneficial in practice. By analyzing the three constructs collectively, the present study reveals trade-offs that single-construct designs risk overlooking. This is exactly the kind of analysis the three-construct framing was designed to enable, and the present findings demonstrate its value.

Secondary analysis: Difficulty. The secondary analysis revealed that the effect of AI support varied substantially across difficulty levels. On Easy phishing emails, both groups performed above 90%, leaving little room for AI to contribute. On Medium and Hard phishing emails, however, the difference was pronounced: control participants classified these correctly at rates similar to or below chance (39% and 47%), while AI-supported participants achieved 70% and 61% respectively. In other words, AI support did not improve performance equally across all difficulty levels. Instead, the largest improvements appeared in the categories where participants without AI performed close to chance. This pattern provides internal evidence for the interpretation presented in the RQ1 discussion: the accuracy gain from AI support was driven primarily by improved performance on difficult phishing emails, where unaided judgment was least reliable, rather than by a uniform improvement across all stimulus types. Given the exploratory nature of this analysis and the number of comparisons conducted, however, these patterns should be interpreted with caution.

7.1 Sustainability

Beyond the methodological and theoretical contributions, the findings of this study also have implications for organizational sustainability. Specifically, successful phishing attacks can cause financial, operational, or reputational damage to organizations. Thus, using decision support tools that reduce the likelihood of successful attacks directly helps build workplace resilience. Additionally, the findings on AI-induced errors presented in this study highlight the need for careful evaluation before any form of deployment. Both considerations align with UN Sustainable Development Goals (specifically SDG 8 and SDG 9), both of which depend on organizations being able to maintain secure and reliable operations. The results also have an educational dimension. By using AI as decision support in phishing detection, users may gradually build their own detection skills, ultimately becoming better equipped to handle threats independently. This dimension aligns with SDG 4 (Quality Education), particularly the targets around lifelong learning and acquiring the skills needed to function in technologically evolving environments. However, AI-supported learning has a critical counterpart: *deskilling*. People could risk becoming too dependent on the AI decision support and their own skills may decline over time if not used responsibly.

8 LIMITATIONS

Group assignment. Due to limitations of the survey tool, participants were assigned to conditions based on whether their birth month was odd or even. While this method is not considered true randomization, both conditions followed the same assignment rule, making it quasi-random. Unlike true randomization, this approach cannot guarantee that the two groups were equivalent on unmeasured factors such as AI literacy or prior phishing exposure, which could introduce some variation between conditions. Birth month, however, has no clear theoretical connection to phishing detection ability, and is therefore unlikely to bias the results.

Sample Size and Sampling Approach. The final sample of 81 participants (approximately 40 per condition) fell short of the a priori target of 128. The smaller sample reduces statistical power, which means small or moderate effects may not have been detected. This is particularly relevant for the secondary confidence analyses (RQ2), where descriptive trends did not reach statistical significance and may reflect underpowering rather than the absence of true effects. The sampling strategy combined purposive, snowball, and convenience sampling, which enabled efficient recruitment across multiple organizations and professional networks, but introduces a degree of selection bias. Participants were primarily working professionals from these networks, who likely have higher digital literacy and email familiarity than a general-population sample. The study did not, however, restrict by role, title, or seniority, allowing for some diversity within the working-professional range. This sampling choice was intentional, as the study targeted individuals who regularly use email in professional settings, reflecting the real-world context of AI-supported phishing detection.

Stimulus Order. Using a fixed stimulus order across all participants could introduce learning effects or fatigue, with later trials potentially showing reduced engagement compared to earlier trials. To reduce this risk, the stimuli were intentionally varied in terms of content, phishing cues, and structure throughout the task. At the same time, since both conditions received the stimuli in the same order, any potential order effects would likely have affected both groups similarly.

9 FUTURE WORK

One natural extension of the present study would be to test the length-bias hypothesis directly. [16] identified explanation length as a key driver of AI-induced confidence inflation, an effect the present study did not replicate, plausibly because the AI explanations were kept deliberately brief. Future work could manipulate explanation length across conditions and examine its effect on user confidence. If confirmed, this would clarify whether the absence of confidence inflation in the present study was driven by the brief explanation format, and would provide stronger evidence that explanation length functions as an important design parameter for AI-supported decision-making systems.

Furthermore, applying a similar research design to other specific high-stakes decisions could generate useful insights on whether the findings of the present study generalize to other domains. In particular, the partial-but-insufficient calibration pattern (where participants override incorrect AI advice some of the time but not enough to prevent errors) could be tested in domains such as medical decision-making, financial decisions, or other contexts where AI-induced errors carry significant consequences. Such extensions would clarify whether the partial-but-insufficient calibration pattern identified here is specific to phishing detection or reflects a broader feature of AI-supported decision-making in high-stakes contexts.

Another potential direction is to examine whether demographic moderators such as AI literacy, prior phishing-detection experience, and domain expertise affect accuracy, confidence, or reliance, since these factors may influence how users weigh their own judgment against AI-generated recommendations. Future work could collect these variables and examine their interaction with the three constructs, clarifying whether the present findings apply uniformly across users or if they differ by background.

10 CONCLUSION

This study examined how generative AI-based decision support affects performance, confidence, and reliance in a risk-sensitive classification task (phishing email detection).

The results of RQ1 showed that AI support improved overall accuracy, primarily by reducing false negatives. The analysis of RQ2 found no detectable change in subjective confidence between conditions, contrary to prior work suggesting AI inflates user confidence. The analysis of RQ3 revealed a pattern of partial-but-insufficient calibration: participants followed incorrect AI advice less often than correct advice, but not often enough to prevent AI-induced overreliance on incorrect recommendations.

Beyond the specific findings, this study contributes to the literature on AI-supported decision-making in two ways. First, by analyzing three constructs jointly on the same task, this study revealed relevant trade-offs that single-construct studies may fail to capture. Second, the seeded-error baseline comparison allowed a more direct assessment of AI-induced error independently of task difficulty.

Overall, these findings suggest that AI decision support does not have one uniform effect on decision-making, but instead affects performance, confidence, and reliance behavior differently. This implies that the net effect of AI support depends on the context in which it is applied. In cybersecurity and similar high-stakes contexts, where false negatives may carry greater costs than false positives, the reduction in false negatives might outweigh the additional errors caused by AI-induced overreliance. More broadly, the findings highlight the importance of evaluating AI-supported decision-making across multiple dimensions rather than relying on a single outcome metric.

ACKNOWLEDGMENTS

A special thank you to the participating companies for your interest, collaboration, and willingness to support the research process. This study would not have been possible without you. I would also like to thank my supervisor for the valuable guidance and feedback provided throughout the project. Lastly, thank you to KTH Royal Institute of Technology for two rewarding years of learning and personal development.

Disclosure of AI use. Generative AI tools, specifically Claude (Anthropic) and ChatGPT (OpenAI), were used for brainstorming, structural feedback, language editing, and for discussing and refining drafts of written material. AI was also used to clarify academic writing conventions and expectations related to master's thesis structure, scope, and presentation, as well as to improve the clarity and readability of parts of the written text. During the research process, generative AI was at times used for clarifying methodological alternatives, statistical procedures, and assisting with implementation of statistical analyses. However, all research design decisions, data collection procedures, analysis strategies, and interpretations of results were made by the author. No sources or references were generated by AI. All cited material was manually identified, reviewed, and verified by the author.

As stated in Section 4.4.2, ChatGPT (GPT-5.3, OpenAI) was used as part of the study's experimental design to generate the AI recommendations for the stimuli in the AI-supported condition.

REFERENCES

[1] Saar Alon-Barkat and Madalina Busuioc. 2023. Human–AI interactions in public sector decision making: "automation bias" and "selective adherence" to algorithmic advice. *J. Public Adm. Res. Theory* 33, 1 (Jan. 2023), 153–169.

[2] Theo Araujo, Natali Helberger, Sanne Kruikemeier, and Claes H de Vreese. 2020. In AI we trust? Perceptions about automated decision-making by artificial intelligence. *AI Soc.* 35, 3 (Sept. 2020), 611–623.

[3] Zana Buçinca, Maja Barbara Malaya, and Krzysztof Z. Gajos. 2021. To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-assisted Decision-making. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW1, Article 188 (April 2021), 21 pages. https://doi.org/10.1145/3449287

[4] Eason Chen, Jeffrey Li, Scarlett Huang, Xinyi Tang, Jionghao Lin, Paulo Carvalho, and Kenneth Koedinger. 2025. AI Knows Best? The Paradox of Expertise, AI-Reliance, and Performance in Educational Tutoring Decision-Making Tasks. (09 2025), 1–8. https://doi.org/10.48550/arXiv.2509.16772

[5] Larry Grullon-Polanco. 2026. LLM-assisted genre analysis: The effect of sampling temperature on reliability. *Research Methods in Applied Linguistics* 5, 2 (2026), 100314. https://doi.org/10.1016/j.rmal.2026.100314

[6] Jeong Hyun Kim, Jungkeun Kim, Seongseop (Sam) Kim, and Tadesse Bekele Hailu. 2023. Effects of AI ChatGPT on travelers' travel decision-making. *Tourism Review* 79, 5 (11 2023), 1038–1057. https://doi.org/10.1108/TR-07-2023-0489 arXiv:https://www.emerald.com/tr/article-pdf/79/5/1038/9564874/tr-07-2023-0489.pdf

[7] Artur Klingbeil, Cassandra Grützner, and Philipp Schreck. 2024. Trust and reliance on AI — An experimental study on the extent and costs of overreliance on AI. *Computers in Human Behavior* 160 (2024), 108352. https://doi.org/10.1016/j.chb.2024.108352

[8] Vivian Lai, Chacha Chen, Q. Vera Liao, Alison Smith-Renner, and Chenhao Tan. 2021. Towards a Science of Human-AI Decision Making: A Survey of Empirical Studies. arXiv:2112.11471 [cs.AI] https://arxiv.org/abs/2112.11471

[9] Siru Liu, Aileen P Wright, Barron L Patterson, Jonathan P Wanderer, Robert W Turer, Scott D Nelson, Allison B McCoy, Dean F Sittig, and Adam Wright. 2023. Using AI-generated suggestions from ChatGPT to optimize clinical decision support. *Journal of the American Medical Informatics Association* 30, 7 (04 2023), 1237–1245. https://doi.org/10.1093/jamia/ocad072 arXiv:https://academic.oup.com/jamia/article-pdf/30/7/1237/50634330/ocad072.pdf

[10] Shuai Ma, Xinru Wang, Ying Lei, Chuhan Shi, Ming Yin, and Xiaojuan Ma. 2024. "Are You Really Sure?" Understanding the Effects of Human Self-Confidence Calibration in AI-Assisted Decision Making. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 840, 20 pages. 9798400703300 https://doi.org/10.1145/3613904.3642671

[11] Marinel Mustata, Dragoş Iliescu, Eugen Mavriş, Cornelia-Rodica Jude, Laviniu Bojor, Paul Tudorache, Ionuţ Cîrdei, Daniela-Elena Hrab, Alina-Cerasela Aluculesei, Ivona Răpan, Stefan-Antonio Dan-Suteu, Daniel Roman, and Constantin Urseiu. 2025. ChatGPT-Assisted Decision-Making: An In-Depth Exploration of the Human–AI Interaction. *International Journal of Human-Computer Interaction* 41 (05 2025), 1–16. https://doi.org/10.1080/10447318.2025.2499165

[12] Briony J Oates. 2006. *Researching Information Systems and Computing*. SAGE Publications, London. 1-4129-0223-1

[13] Irdin Pekaric, Philipp Zech, and Tom Mattson. 2025. LLMs in Cybersecurity: Friend or Foe in the Human Decision Loop? arXiv:2509.06595 [cs.CR] https://arxiv.org/abs/2509.06595

[14] Emely Rosbach, Jonathan Ganz, Jonas Ammeling, Andreas Riener, and Marc Aubreville. 2025. *Automation Bias in AI-assisted Medical Decision-making under Time Pressure in Computational Pathology*. 129–134. 978-3-658-47421-8 https://doi.org/10.1007/978-3-658-47422-5_27

[15] Yiqiu Shen, Laura Heacock, Jonathan Elias, Keith D Hentel, Beatriu Reig, George Shih, and Linda Moy. 2023. ChatGPT and other large language models are double-edged swords. *Radiology* 307, 2 (April 2023), e230163.

[16] Mark Steyvers, Heliodoro Tejeda, Aakriti Kumar, Catarina Belem, Sheer Karny, Xinyue Hu, Lukas W. Mayer, and Padhraic Smyth. 2025. What large language models know and what people think they know. *NATURE MACHINE INTELLIGENCE* 7, 2 (FEB 2025), 221–231. https://doi.org/10.1038/s42256-024-00976-7

[17] Francisco Teixeira-Marques, Nuno Medeiros, Francisco Nazaré, Sandra Alves, Nuno Lima, Leandro Ribeiro, Rita Gama, and Pedro Oliveira. 2024. Exploring the role of ChatGPT in clinical decision-making in otorhinolaryngology: a ChatGPT designed study. *Eur. Arch. Otorhinolaryngol.* 281, 4 (April 2024), 2023–2030.

[18] Michelle Vaccaro, Abdullah Almaatouq, and Thomas Malone. 2024. When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nat. Hum. Behav.* 8, 12 (Dec. 2024), 2293–2303.

A AI PROMPT

The following prompt structure was used in ChatGPT (GPT-5.3) to generate AI recommendations for each stimulus email.

System prompt:

"Classify the following email as either phishing or legitimate and provide a brief explanation (1–2 sentences) for the classification. Do not include numeric confidence estimates."

User prompt:

"[EMAIL CONTENT]"

For the six seeded-error trials, the AI classification was manually replaced with the opposite classification, as described in Section 4.4.1.

B EXAMPLE EMAIL STIMULI AND AI RECOMMENDATIONS

The following examples illustrate representative stimuli used in the experiment across different difficulty categories. The examples also demonstrate the format of the AI-generated recommendations shown to participants in the AI-supported condition.

B.1 Example 1: Easy Phishing Email

Difficulty category: Easy phishing
Ground truth: Phishing
Seeded error: No

Email stimulus

> From: support@account-security-alerts.com
> To: You
> Subject: Urgent: Verify your account immediately
>
> We have detected a critical issue with your account.
> To avoid suspension, you must verify your account details immediately.
>
> Please sign in below to confirm your information:
> [Sign in]
> Failure to complete this process may result in limited account access.

AI recommendation shown to participants

> Classification: Phishing
> Explanation: The email pressures you to act immediately and asks you to verify your account through a link.

B.2 Example 2: Medium Phishing Email

Difficulty category: Medium phishing
Ground truth: Phishing
Seeded error: No

Email stimulus

> From: no-reply@docusign.com
> To: You
> Subject: Signature request: Agreement_2026_v2
>
> You have received a document for signature:
>
> Agreement_2026_v2
>
> Please review and complete the signing process below:

[Review & Sign]

Thanks,
Microsoft Document Services

AI recommendation shown to participants

Classification: Phishing
Explanation: The email mixes DocuSign with “Microsoft Document Services,” which is inconsistent.

B.3 Example 3: Hard Phishing Email

Difficulty category: Hard phishing
Ground truth: Phishing
Seeded error: No

Email stimulus

From: account-noreply@microsoft.com
To: You
Subject: New sign-in detected

We detected a sign-in to your Microsoft account from a new device earlier today.

If this was you, no action is required. To confirm this was you and secure your account, please verify your identity below:

[Verify identity]
Microsoft Account Team

AI recommendation shown to participants

Classification: Phishing
Explanation: The email gives inconsistent guidance about the next step, which is unusual for a legitimate security notice.

B.4 Example 4: Clear Legitimate Email

Difficulty category: Clear
Ground truth: Legitimate
Seeded error: No

Email stimulus

From: hr@company.com
To: You
Subject: Friday breakfast

Hi,

We’ll be hosting a breakfast in the office kitchen this Friday at 09:00.
Feel free to drop by.

Best,
HR Team

AI recommendation shown to participants

Classification: Legitimate
Explanation: The email is a simple workplace announcement about a breakfast event.

B.5 Example 5: Ambiguous Legitimate Email

Some examples included intentionally incorrect AI recommendations (“seeded errors”) used to examine reliance on faulty AI outputs. An example of this is included below.

Difficulty category: Ambiguous
Ground truth: Legitimate
Seeded error: Yes

Email stimulus

From: johan.lindstrom@company.com
To: You
Subject: Quick question

Hi,

I just shared a SharePoint document with you - can you check if you’re able to access it?

[Open document]

AI recommendation shown to participants

Classification: Phishing
Explanation: The email asks you to follow a document link without clearly indicating what the document contains, which can be suspicious.

TRITA-EECS-EX-2026:293
Stockholm, Sweden 2026

www.kth.se